

S. Sandra Bae

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RESEARCH INTEREST

My research spans across human-computer interaction (HCI), data visualization, and computational fabrication. I study the interplay of the digital and physical worlds when people analyze high-dimensional and spatial data.

This intersection offers a uniquely reciprocal space where scientific inquiry and engineering invention co-evolve: to understand how people think with data through multiple senses, we must also engineer the materials and mechanisms that make such thinking possible. My research advances the *science* of data physicalization (i.e., physical representations of data) by uncovering how humans perceive, interpret, and reflect on information when it is made tangible—probing the perceptual, tactile, and metacognitive processes that govern data sensemaking. In parallel, it advances the *engineering* of physicalization by developing scalable, perceptually coherent representation systems that integrate computation, fabrication, sensing, and material behavior to support real analytical work.

Keywords: Human-Computer Interaction, Data Visualization, Computational Geometry & Fabrication, Optimization, Tangible Interactions

Last Updated (Feb 2026)

EMPLOYMENT

2025 - PRESENT **University of Arizona**
Assistant Professor
Department of Computer Science

2025 - PRESENT **University of Arizona**
Faculty Affiliate
Cognitive Science

EDUCATION

2020 - 2025 **University of Colorado, Boulder**
Ph.D. in Creative Technology & Design
Thesis: Advancing Interactive Physicalizations

2018 - 2020 **University of California, Davis**
M.S. in Computer Science

2014 - 2018 **University of California, Davis**
B.A. in Human-Computer Interaction
Minor in Education

AWARDS AND DISTINCTION

2025 **Best Paper (ACM CHI25) ***
2024 **APUP Researcher at the National Renewable Energy Laboratory (\$230,000) ***
2024 **KASF-KIA Scholarship (\$2,500) (Top 5 recipient for KIA special recognition for excellence in STEM) ***
2024 **CU Boulder's Special Recognition Award for Partnership for Sustainability (*Coffee Ground Filament*)**
2023 **EECS Rising Star ***
2023 **Best Paper Honorable Mention (IEEE VIS23) ***
2023 **Korean American Scholarship Foundation (\$2,000)**
2023 **Achievement Reward for College Scientists (\$7,500)**
2023 **Rocky Mountain RepRap Festival, Most Functional/Useful, Sensing Networks ***

2023 Rocky Mountain RepRap Festival, Most Innovative, *Coffee Ground Filament*
 2022 Achievement Reward for College Scientists (\$7,500)
 2022 Korean American Scholarship Foundation (\$3,000)
 2022 David T. Spalding Graduate Teaching Fund Fellowship (\$1000)
 2021 Korean American Scholarship Foundation (\$2,000)
 2021 CU Boulder Travel Grant (\$500)
 2021 CU Boulder Dean's Engineering Travel Grant (\$400)
 2021 Honorable Mention, IEEE World Haptics Student Design Competition
 2021 CRA-WP Grad Cohort for Women
 2021 Ada Lovelace Fellow, Open Hardware Summit
 2021 Craft Award, ACM TEI Student Design Competition.
 2021 Achievement Reward for College Scientists (\$6,500)
 2019 Richard C. and Joy Dorf Engineering Graduate Fellowship (\$1,000)
 2019 **NASA JPL Master's Thesis Educational Fellow (\$30,000) ***
 2018 NSF's Preparing Engineering Graduate Students for the 21st Century Fellow (\$10,000).
 2018 Honorable Mention Visual Storytelling Award, IEEE Pacific Visualization Symposium.
 2017 Beneath (CONNECT EXPO 2017) - Judge's Honorable Mention.
 2017 UC Davis Organizational Research Expo.
 2017 Better Together - AIGA and IDSA.

RESEARCH EXPERIENCE

WINTER 2024 - SUMMER 2025	National Renewable Energy Lab (NREL) Golden, CO <i>Graduate Research Assistant</i> Research with Kenny Gruchalla, Olga Doronina, and Ryan King. Developing an analytical tangible user interface to manipulate affine transformations for aerodynamic design (e.g., wind blades). In collaboration with NIST (Dr. Zachary Grey).
FALL 2020 - SUMMER 2025	University of Colorado, Boulder Boulder, CO <i>Graduate Research Assistant & Teaching Assistant</i> Research with Danielle Albers Szafir, Ellen Yi-Luen Do, and Michael L. Rivera. Teaching assistant for Daniel Leithinger, Laura Devendorf in ATLAS.
SUMMER 2020	Stanford University Palo Alto, CA (Remote) <i>HCI Research Intern</i> Research with James Landay and Elizabeth Murnane. Identifying and building family-centered, in-car technology to support collaborative learning.
SUMMER 2019 - SUMMER 2020	NASA Jet Propulsion Lab Pasadena, CA <i>University Researcher</i> Research with Federico Rossi, Scott Davidoff, and Joshua Vander Hook. Working as a NASA JPL Master's Thesis Fellow to expand Summer 2019's project.
SUMMER 2019	NASA Jet Propulsion Lab Pasadena, CA <i>Data Visualization / HCI Research Intern</i> Research with Hillary Mushkin, Santiago Lambedya, Maggie Hendrie, Scott Davidoff. Developed a visual analytics systems to analyze scheduling coordination for multi-robot systems
SUMMER 2019 - SPRING 2020	University of California, Davis Davis, CA <i>Undergraduate / Graduate Student Research Assistant & Teaching Assistant</i> Research with Kwan-Liu Ma. Teaching assistant for Kwan-Liu Ma, Hao-Chuan Wang, and Nina Amenta in Computer Science.

REFEREED PUBLICATIONS

I author in both **computer science** and **physical science** venues. For *physical sciences*, I author in journals, such as the Journal of Astronautical Sciences. In *computer science*, conferences are the primary venues (~25% acceptance rates). I author in ACM CHI Conference on Human Factors in Computing Systems (CHI), the ACM conference on Designing Interactive Systems (DIS), IEEE Visualization Conference (IEEE VIS), and IEEE International Conference on Robotics and Automation (IEEE ICRA).

JOURNAL ARTICLES

- J1 S. S. Bae, K. Cave, C. Görg, P. Rosen, D. A. Szafir, and C. X. Bearfield. “Bridging Network Science and Vision Science: Mapping Perceptual Mechanisms to Network Visualization Tasks”. In: *IEEE Transactions on Visualization and Computer Graphics* (2025).
- J2 S. S. Bae, T. Fujiwara, A. Ynnerman, E. Y.-L. Do, M. Rivera, and D. A. Szafir. “A Computational Design Pipeline to Fabricate Sensing Network Physicalizations”. In: *IEEE Transactions on Visualization and Computer Graphics (also proc. IEEE VIS 2023)*. Melbourne, Australia, 2023, Acceptance rate: 24.7% (133/539)
* **Best Paper Honorable Mention (Top 3%)**.
- J3 S. S. Bae, R. Vanukuru, R. Yang, P. Gyory, R. Zhou, E. Y.-L. Do, and D. A. Szafir. “Cultivating Visualization Literacy for Children Through Curiosity and Play”. In: *IEEE Transactions on Visualization and Computer Graphics (also proc. IEEE VIS 2022)*. 2022, Acceptance rate: 26.5% (122/460).
- J4 R. Woollands, F. Rossi, T. S. Vaquero, M. S. Net, S. S. Bae, V. Bickel, and J. V. Hook. “Maximizing Dust Devil Follow-Up Observations on Mars Using Cubesats and On-board Scheduling”. In: *Journal of Astronautical Sciences*. 2022.

CONFERENCE ARTICLES

- C1 K. Ranjan, S. S. Bae, P. Gyory, E. Y.-L. Do, R.-H. Liang, and C. Zheng. “Why (not) ReacTiVision: Charting Emerging Challenges and Opportunities for Building Tangible User Interfaces with Computer Vision”. In: *Proceedings of the Twentieth International Conference on Tangible, Embedded, and Embodied Interaction (TEI)*. Chicago, USA, 2026, Acceptance rate: 22.65% (41/181).
- C2 S. S. Bae*, T. Fujiwara*, D. A. Szafir, E. Y.-L. Do, and M. L. Rivera. “Computational Design and Single-Wire Sensing of 3D Printed Objects with Integrated Capacitive Touchpoints”. In: *Proceedings of the 10th ACM Symposium on Computational Fabrication*. Boston, USA, 2025, (*equally contributed).
- C3 S. S. Bae*, T. Fujiwara*, C. Tseng*, and D. A. Szafir. “Uncovering How Scatterplot Features Skew Visual Class Separation”. In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. Yokohama, Japan, 2025, Acceptance rate: 25.1% (1249/5020).
- C4 C. Kohlbrenner, C. Esobedo, S. S. Bae, A. Dickhans, and A. Roncone. “GenTact Toolbox: A Computational Design Pipeline to Procedurally Generate Context-Driven 3D Printed Large-Area Tactile Skins”. In: *Proceedings of the 2025 IEEE International Conference on Robotics and Automation*. Atlanta, Georgia, 2025, Acceptance rate: 38.67% (1606/4153).
- C5 R. Sorenson-Graff, S. S. Bae, and J. Wirfs-Brock. “Integrating Annotations for Sonifications and Physicalizations”. In: *Proceedings of the IEEE VIS 2024*. St. Pete Beach, Florida, 2024, Acceptance rate: 31.9% (66/207).
- C6 X. Wen, S. S. Bae, and M. L. Rivera. “Enabling Recycling of Multi-Material 3D Printed Objects through Computational Design and Disassembly by Dissolution”. In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. Yokohama, Japan, 2024, Acceptance rate: 25.1% (1249/5020)
* **Best Paper (Top 1%)**.
- C7 P. Gyory, S. S. Bae, R. Yang, E. Y.-L. Do, and C. Zheng. “Marking Material Interactions with Computer Vision”. In: *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. Hamburg, Germany, 2023, Acceptance rate: 28.39% (880/3182).
- C8 M. Rivera, S. S. Bae, and S. Hudson. “Exploring Spent Coffee Grounds as a Sustainable Material for Prototyping with 3D Printing”. In: *Proceedings of the 2023 on Designing Interactive Systems Conference*. Pittsburg, PA, USA, 2023, Acceptance rate: 24.1% (174/726).
- C9 S. S. Bae, D. A. Szafir, and E. Y.-L. Do. “Exploring the Benefits and Challenges of Data Physicalization”. In: *Proceedings of the Fourth European Tangible Interaction Studio (ETIS’ 22)*. Toulouse, France, 2022, Acceptance rate: 88% (15/17).

- C10 S. S. Bae, C. Zheng, M. E. West, E. Y.-L. Do, S. Huron, and D. A. Szafrir. “Making Data Tangible: A Cross-disciplinary Design Space for Data Physicalization”. In: *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. New Orleans, Louisiana, 2022, Acceptance rate: 23% (4590/19964).
- C11 S. S. Bae, F. Rossi, J. V. Hook, S. Davidoff, and K.-L. Ma. “A Visual Analytics Approach to Debugging Cooperative, Autonomous Multi-Robot Systems’ Worldviews”. In: *Proceedings of the IEEE Visual Analytics Science and Technology (VAST) 2020*. Salt Lake City, Utah (Virtual), 2020, Acceptance rate: 24.8% (52/210).
- C12 S. S. Bae*, O.-H. Kwon*, S. Chandrasegaran, and K.-L. Ma. “Spinneret: Aiding Creative Ideation through Non-Obvious Concept Associations”. In: *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. Honolulu, Hawaii (Virtual), 2020, (*equally contributed), Acceptance rate: 24.31% (760/3126).

DEMOS & POSTERS

- D1 R.-H. Liang, S. Houben, K. Ranjan, S. S. Bae, P. Gyory, E. Y.-L. Do, and C. Zheng. “ArUcoTUI: Software Toolkit for Prototyping Tangible Interactions on Portable Flat-Panel Displays with OpenCV”. In: *Proceedings of the Twentieth International Conference on Tangible, Embedded, and Embodied Interaction (TEI)*. Chicago, USA, 2026.
- D2 S. S. Bae*, T. Fujiwara*, D. A. Szafrir, E. Y.-L. Do, and M. L. Rivera. “Demonstration of Computational Design and Single-Wire Sensing of 3D Printed Objects with Integrated Capacitive Touchpoints”. In: *Proceedings of the 10th ACM Symposium on Computational Fabrication*. Boston, USA, 2025, (*equally contributed).
- D3 S.-Y. Ma, S. S. Bae, E. Y.-L. Do, T. Yeh, and E. Bradley. “From Flat Screens to Immersive Space: Rethinking Multimodal Learning of Complex Systems”. In: *Proceedings of the Eleventh Taiwan Human-Computer Interaction Symposium (TAICHI ’25)*. Hsinchu, Taiwan, 2025.
- D4 X. Wang, H. C. van Iterson, S. S. Bae, and R.-H. Liang. “Reflecting on Solo Dining Behavior with Annotated Data Physicalization”. In: *Proceedings of the Eighteenth International Conference on Tangible Embedded and Embodied Interaction (TEI ’25)* (2025).
- D5 S. S. Bae, T. Fujiwara, D. A. Szafrir, E. Y.-L. Do, and M. L. Rivera. “3D printed multi-point capacitive touch sensor”. In: *Symposium on Computational Fabrication*. SCF ’24. Aarhus, Denmark: Association for Computing Machinery, 2024.
- D6 E. Johnson, S. S. Bae, and E. Y.-L. Do. “Supporting Data Visualization Literacy through Embodied Interactions”. In: *Proceedings of the Fifteenth International Conference on Creativity & Cognition (C&C ’22)*. Gathertown (Remote): Association for Computing Machinery, 2023.
- D7 S. S. Bae. “Towards a Deeper Understanding of Data and Materiality”. In: *Proceedings of the Fourteenth International Conference on Creativity & Cognition (C&C ’22)*. Venice, Italy: Association for Computing Machinery, 2022.
- D8 S. S. Bae, R. Yang, P. Gyory, J. Uhr, D. A. Szafrir, and E. Y.-L. Do. “Touching Information with DIY Paper Charts & AR Markers”. In: *Interaction Design and Children*. IDC ’21. Athens, Greece (Virtual): Association for Computing Machinery, 2021, Acceptance rate: 27% (136/501).
- D9 S. S. Bae* and M. E. West*. “Cyborg Crafts: Second SKIN (Soft Keen INteraction)”. In: *Proceedings of the Fifteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI ’21)*. Salzburg, Austria (Virtual): Association for Computing Machinery, 2021, (*equally contributed).

TECH REPORTS

- T1 R. Woollands, F. Rossi, T. S. Vaquero, M. S. Net, S. S. Bae, V. Bickel, and J. V. Hook. “Maximizing Dust Devil Follow-Up Observations on Mars Using Cubesats and On-board Scheduling”. In: *Proceedings of the 43rd Annual AAS Guidance & Control Conference, Breckenridge, CO*. 2021.

BOOK CHAPTERS

- B1 T. Hopkins, S. S. Bae, J. Uhr, C. Zheng, A. Banic, and E. Y.-L. Do. “User Interfaces in Smart Cities”. In: *Handbook of Smart Cities*. Ed. by J. C. Augusto. Springer International Publishing, 2021.

PATENT AND INDUSTRY PARTNERSHIP

NON-PROVISIONAL PATENT APPLICATION

Bae, S. S, Fujiwara, T., Rivera, M.L., Szafir, D.A., Do., E.Y.D (2023). Systems and Methods for Touching Sensing Based on Resistor-Capacitor Delays. *U.S. Patent Application 63/497,594*, filed April 21, 2023. Pending non-provisional patent application.

PROVISIONAL PATENT APPLICATION

Bae, S. S, Fujiwara, T (2025). Systems and Fabrication Methods for Resistor Designs Through Printing. *UA26-100P*, filed November 16, 2025.

INDUSTRY PARTNERSHIP

Company	Funding	Year
Stratasys	Gift	2024
Protopasta	Sponsored Research	2024, 2025

TALKS

All first-author conference papers listed above were also given as presentations at their respective conferences and are not listed again in this section.

INVITED GUEST LECTURER

Escaping Flatland: Data Physicalization Now, Multi-Modal Agents Next

Seoul National University, *Information Visualization and Visual Analytics*, December 7, 2025 (hosted by Jinwook Seo)

Expanding your Interactive Toolkit: Computer Vision Markers and 3D Printed Capacitive Sensors

Georgia Tech, *Prototyping Interactive Systems*, November 7, 2023 (hosted by Hyunjoo Oh)

Using Computational Fabrication for Data Physicalization

UNC-Chapel Hill, *Information Professionals in the Makerspace*, October 17, 2023 (hosted by Maggie Melo)

Computational Fabrication for Electronics

CU Boulder, *Computational Fabrication*, October 16, 2023 (hosted by Michael L. Rivera)

The Value of Design: Empowering Humans through Human-Centered Thinking

CU Boulder, *Information Visualization*, March 18, 2021 (hosted by Danielle Szafir)

The Value of Design: Empowering Humans through Human-Centered Thinking

UC Davis, *Android Development*, January 14, 2021 (hosted by Nina Amenta)

INVITED TALKS AND PRESENTATIONS

Communicating Complex Data through 3D Data Representation

University of Arizona, February 9, 2026 (hosted by Swarna Garnesh)

Escaping Flatland: Data Physicalization Now, Multi-Modal Agents Next

University of Utah, December 3, 2025 (hosted by Andrew McNutt)

AI, Meet Matter: How AI Generates Forms, Mediates Interactions, and Models Cognition

University of Arizona AI Club, November 12, 2025 (hosted by Zhuolin Yang)

IEEE VIS 2025 Doctoral Colloquium

Panelist, IEEE VIS 2025 Doctoral Colloquium

Escaping Flatland: 3D+ Data Visualizations for New Users, Interactions, and Analytical Environments

TU Wien, October 31, 2025 (hosted by Renata Raidou)

Simple Tools, Creative Outcomes: A Computer Vision-Based Approach

Computer Science Teachers Organization (CSTA-Arizona), September 11, 2025 (hosted by Mihai Surdeanu)

Escaping Flatland: 3D+ Data Visualizations for New Users, Interactions, and Analytical Environments

University of Texas, Dallas (UT Dallas)

University of Arizona

Worcester Polytech Institute (WPI)

Washington University in St. Louis

Tufts University

Purdue University

Lifting Visualizations Beyond Flatland: A Geometric Perspective

National Institute of Science and Technology (NIST), October 1, 2024 (hosted by Matt Whitlock)

Pushing Visualizations Beyond the Desktop with Everyday Devices and Novel Interfaces

University of Iowa, November 17, 2023 (hosted by Juan Pablo Hourcade)

Pushing Visualizations Beyond the Desktop with Everyday Devices and Novel Interfaces

Emory University, November 8, 2023 (hosted by Emily Wall)

Pushing Visualizations Beyond the Desktop with Everyday Devices and Novel Interfaces

Georgia Tech, November 8, 2023 (hosted by Yalong Yang)

Using Network Science to Reimagine Everyday Objects as Future Computers

TTI Vanguard, September 13, 2023 (hosted by Nancy Kleinrock)

Towards “Best of Both Worlds”: Bridging Digital and Physical Representations in Visualization Research

Linköping University, January 9, 2023 (hosted by Miriah Meyer)

Debugging Multi-Robot Autonomous System Anomalies

CU Boulder ATLAS Seminar, October 6, 2020 (hosted by Ellen Yi-Luen Do)

Debugging Multi-Robot Autonomous System Anomalies

NASA-JPL, August 27, 2020 (hosted by Scott Davidoff)

TEACHING**TEACHING EXPERIENCE**** indicates the instructor of the course*

Term	Institute	Subject Title	Role	Enrolled	Evals
SPRING 2026	UArizona	CSC696C: Compute, Fabricate, Interact	Instructor	6	No
FALL 2025	University of Washington	3D Printing (Brave Behind Bars ¹)	Teaching assistance Martin Nisser*	12	No
SUMMER 2024	MIT	Web Development (Brave Behind Bars ¹)	Teaching assistance Thomer Gil*	54	No
SPRING 2022	CU Boulder	ATLAS3300: Object (Physical Computing)	Teaching assistance Laura Devendorf*	37	No
SPRING 2021	CU Boulder	ATLAS3300: Object (Physical Computing)	Teaching assistance Daniel Leithinger*	36	No

¹Brave Behind Bars is a college-accredited introductory computer science and career-readiness program for incarcerated people.
<https://bravebehindbars.org/>

SPRING 2020	UC Davis	ECS162: Web Programming	Teaching assistance Nina Amenta*	122	Yes
WINTER 2020	UC Davis	ECS164: Intro to Human- Computer Interaction	Teaching assistance Hao-Chuan Wang*	59	Yes
WINTER 2019	UC Davis	ECS164: Intro to Human- Computer Interaction	Teaching assistance Hao-Chuan Wang*	63	Yes
SUMMER 2018	UC Davis	ECS163: Information Visualization	Teaching assistance Kwan-Liu Ma*	23	No

ADVISING & MENTORSHIP

PH.D. STUDENT (SUPERVISED)

Name	Degree	Year
Yanyan Dong	Ph.D. in Information Science (University of Arizona) * co-advised with Steve Bethard	2025 – PRESENT

PH.D. STUDENT (RESEARCH PROJECT MENTOR)

Name	Degree	Year
Krithik Ranjan	Ph.D. in Creative, Tech. & Design (CU Boulder)	2024 – 2025
Xin Wen	Ph.D. in Creative, Tech. & Design (CU Boulder)	2024
Chin Tseng	Ph.D in Computer Science (UNC-Chapel Hill)	2023–2024
Peter Gyory	Ph.D. in Creative, Tech. & Design (CU Boulder)	2022–2023

MASTER STUDENTS

Name	Degree	Year
Quintan Gerhardstein	M.S. in Creative, Tech. & Design (CU Boulder)	2024
Ada Zhao	M.S. in Creative, Tech. & Design (CU Boulder)	2024
Carson Kohlbrenner	M.S. in Aerospace Engineering (CU Boulder)	2023–2024
Ashleigh Thomas	M.S. in Computer Science (UC Davis)	2019

UNDERGRADUATE STUDENTS

Name	Degree	Year
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Jordan Orvik	B.S. in Computer Science (University of Arizona)	2025 – PRESENT
Gwendolyn Alviar	B.S. in Mechanical Engineering (University of Arizona)	2025 – PRESENT
William Bettis	B.A. in Design Arts & Practices (University of Arizona)	2025 – PRESENT
Noah Liska	B.S. in Computer Science (CU Boulder)	2023–2024
Caleb Kumar	B.S. in Computer Science (CU Boulder)	2023–2024
Rhys Sorenson-Graff	B.A. in Computer Science (Whitmann College)	2023–2024
Eloise Yalovitser	B.FA in Design & Tech. (Parson School of Design)	2023
Antonio (Zhixing) Li	B.S. in Mechanical Engineering (CU Boulder)	2023
Elise Johnson	B.S. in Mechanical Engineering (CU Boulder)	2021–2022
Vy Thai	B.S. in Computer Science (Stanford University)	2020
Lovpret Kaur	B.S. in Computer Science (UC Davis)	2019

SUPERVISED THESES

	Total	Completed	In Progress
Doctor of Philosophy	2	0	2
As Advisor	1	0	1
As Committee	1	0	1
Master of Science (MS)	1	0	1
As Advisor	0	0	0
As Committee	1	0	1
Bachelor of Science (BS)	1	0	1

PhD Theses, Advisor

Name	Title	(Dept.) Year
Yanyan Dong		Expected 2030

PhD Theses, Committee

Name	Title	(Dept.) Year
Tanner A. Finken	Discrete Vector Field Improvement, Supporting Fast Generation, Provable Correctness, and Adaptable Simplification	2026

MS Theses, Committee

Name	Title	(Dept.) Year
Md Afridi Hasan	Analyzing Sensitivity of Visualization-Purpose Dimensionality Reduction Methods to Data Perturbations	2026

BS Theses, Advisor

Name	Title	(Dept.) Year
Jordan M. Orvik	Mediating the Follow: A Longitudinal Study on Reflective Practice and its Effect on Parasocial Relationships on Instagram	2026

ACADEMIC SERVICE TO PROFESSIONAL COMMUNITY

Program Committee

ACM Creativity & Cognition 2021
ACM Tangible and Embedded Interaction (TEI) 2024
IEEE VIS 2026
IEEE Pacific Visualization Conference (Visualization Notes) 2025
EG/VGTC EuroVis (Short Paper) 2026

Organizing Committee

IEEE VIS 2026, 2027 (Inclusivity Chair)
IEEE VIS 2022, 2024 BELIV Workshop (Social Media and Web Chair)

Invited Conference Reviewer

ACM Symposium on User Interface Software and Technology (UIST) 2021
ACM Interaction Design and Children (IDC) 2023
ACM Designing Interactive Systems (DIS) 2021–2024
ACM Conference on Human Factors in Computing Systems (CHI) 2021–2026
ACM Tangible and Embedded Interaction (TEI) 2022–2026
ACM Creative & Cognition (CC) 2021–2022
EG/VGTC EuroVis 2022, 2026
NordicCHI 2024
SIGGRAPH Asia Technical Paper 2024
SIGGRAPH Technical Paper 2025, 2026
IEEE VR 2025

Invited Journal Reviewer

IEEE Transactions on Visualization and Computer Graphics (TVCG) 2022–2025
International Journal of Child-Computer Interaction 2022
Behaviour & Information Technology 2023, 2024
IS&T Journal of Perceptual Imaging 2024, 2025
International Journal of Human-Computer Studies 2025

Invited Book Chapter Reviewer

“Visualization Psychology” published by Springer Nature. Editors: Danielle Albers Szafir, Rita Borgo, Min Chen, Darren J. Edwards, Brian Fisher, & Lace Padilla

Special Recognition as a Reviewer

2023 ACM Tangible and Embedded Interaction (TEI)
2023 ACM Conference on Human Factors in Computing Systems (CHI) x2
2024 ACM Conference on Human Factors in Computing Systems (CHI) x2
2026 ACM Conference on Human Factors in Computing Systems (CHI)

Other Conference Organizational Roles

Faculty Panelist, IEEE VIS 2025 Doctoral Colloquium
Session Chair, IEEE VIS 2025 Full Paper Presentation

ACADEMIC SERVICE TO UNIVERSITY

Departmental Service

Graduate Admissions Committee, Department of Computer Science, 2025–2026
Seminar Chair, Department of Department of Computer Science, 2025–2026
Ph.D. Open House Organizer 2021, ATLAS Institute, CU Boulder
Faculty Candidate Student Host 2021, 2024 ATLAS Institute, CU Boulder
Ph.D. Graduate Application Support Program Founder 2022, 2023, ATLAS Institute, CU Boulder

DESIGN EXHIBITIONS

2019 Sit On Data | Imagining America's 20th Anniversary National Gathering

A data-driven parametric-bench where users can feel the data by sitting. Developing using Python, CNC routing, Laser cutting.

2017 BENEATH | SOFA Chicago 2017

An immersive installation conveying California's land subsidence. Presented at SOFA Chicago 2017. Built motion graphics and projection using Processing.

2017 OneClimate: Prototyping Climate Change | Exploratorium Museum (SF)

A multisensory exhibition conveying the impact of climate change sponsored by the John Muir Institute of the Environment. Featured at the Exploratorium in San Francisco for the OneClimate event, AIGA & ISAD - San Francisco, and IA's 2017 National Gathering. Designed the scientific narrative and constructed the exhibit using woodshop skills.

SELECTED MEDIA COVERAGE

2023 Yahoo Finance | First-of-its-kind study combines common household waste product with 3D printing — here are the results

<https://finance.yahoo.com/news/first-kind-study-combines-common>

2023 Ars Technica | Don't throw out those used coffee grounds—use them for 3D printing instead

<https://arstechnica.com/science/2023/09/dont-throw-out-those-used-coffee-grounds-use-them-for-3d-printing-instead/>

2023 TechBriefs | Using Coffee to Reduce Waste from 3D Printing

<https://www.techbriefs.com/component/content/article/tb/stories/blog/49103>

2023 Daily camera (Boulder) | CU Boulder professor creates 3D printing paste with coffee grounds

<https://www.dailycamera.com/2023/09/20/cu-boulder-professor-creates-3d-printing-paste-with-coffee-grounds/>

2023 CNC Kitchen (Youtube) | 3D Printing Resistors, Fibers & Coffee - Utility Research Lab

https://youtu.be/Oeqvo2c28_c

2023 Make Magazine | Rocky Mountain RepRap Fest: All The Cool 3D Printing Stuff

<https://makezine.com/article/digital-fabrication/3d-printing-workshop/3d-printing-festival-loveland-colorado/>

2023 Meltzone Podcast | Rocky Mountain RepRap Festival & Prusa MK4

<https://youtu.be/NkgOXFn55cM?t=1041>

2023 CU Boulder | ATLAS Innovators Win Big at RepRap Festival

<https://www.colorado.edu/atlas/2023/05/08/atlas-innovators-win-big-reprap-festival>

2023 CU Boulder | New seed grants from the Engineering Education and AI-Augmented Learning research theme will support range of topics

<https://www.colorado.edu/irt/engineering-education-ai/2023/05/19/new-seed-grants-engineering-education-and-ai-augmented-learning-research-theme-will>

2021 CU Boulder | Sandra Bae receives ARCS and KASF scholarship

<https://www.colorado.edu/atlas/2021/09/08/sandra-bae-receives-arcs-and-kasf-scholarships>

2017 CONNECT EXPO | SOFA CHICAGO 2017 Exhibition

Printing Press: Chubb (pg. 10-11).

2017 AIGA and IDSA | Better Together Design Exhibition

Printing Press: AIGA (pg. 8-9).

2017 UC Davis | 'OneClimate' a Call to Arms

<https://www.ucdavis.edu/news/oneclimate-call-arms>